EDUGENIE: GOOGLE GEMINI POWERED LEARNING ASSISTANT

MILESTONE 1: MODEL SELECTION AND ARCHITECTURE

Activity 1.1: Select AI Models

1. Gemini 1.5 Pro (via API)

Used for:

- Question answering
- Summarization
- Quiz generation
- Learning path recommendations
- Document-based learning assistance

Explanation: Gemini 1.5 Pro is a powerful Generative AI model provided through the Google Gemini API. In the EduGenie project, it is used to understand user questions and uploaded learning materials. It can generate summaries, answer questions, create quizzes, and provide personalized learning recommendations.

Benefits:

- Advanced reasoning capability
- Generates structured and meaningful responses
- Suitable for document-based learning
- Supports cloud-based AI inference
- Helps provide personalized learning assistance

2. LaMini-Flan-T5-783M (Local Model)

Used for:

- Concept explanation
- Basic question answering
- Simple educational responses

Explanation: LaMini-Flan-T5-783M is a lightweight instruction-tuned language model that can be used locally. In the EduGenie project, it can be used for providing simple explanations and basic educational responses without depending completely on a cloud-based API.

Benefits:

- Lightweight model
- Instruction-tuned
- Can run locally
- CPU-compatible
- Reduces dependency on cloud services for basic tasks

Folder Architecture

No.	File / Folder	Purpose
1	main.py	Main application
2	explanation_module.py	Concept explanation
3	qna.py	Question answering
4	quiz_module.py	Quiz generation
5	summary_module.py	Summarization
6	learning_path.py	Learning recommendations
7	templates/index.html	HTML frontend
8	static/style.css	Styling
9	requirements.txt	Python dependencies

Project Architecture Explanation

The EduGenie project follows a modular architecture. The main application connects different modules such as question answering, summarization, quiz generation, concept explanation, and learning recommendations.

The user can provide a topic or upload a document. The application processes the input using the selected AI model and generates useful learning content such as notes, answers, quizzes, summaries, and learning recommendations.